Sharp Moment Stability of Brenier Maps

1 Notation and statement

Throughout, $d \geq 2$. Our source is a fixed probability measure

$$\rho = f\,dx, \qquad \operatorname{supp}\rho = K, \tag{1.1}$$

where $K \subset \mathbb{R}^d$ is a compact convex body with nonempty interior and f is log-concave. We take a representative positive and continuous on $\operatorname{int} K$ and zero outside $\operatorname{int} K$; boundary values do not affect the measure. In particular, no positive lower bound for f on all of K is assumed. Log-concavity means

$$f((1-t)x + tz) \geq f(x)^{1-t} f(z)^t \qquad (0 < t < 1).$$

The standard interior continuity of a positive log-concave density follows from continuity of the finite concave function $\log f$.

Let $\mathcal{P}_2(\mathbb{R}^d)$ denote the Borel probability measures with finite second moment. For $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$, let

$$W_2(\mu,\nu)^2 = \inf_{\pi \in \Pi(\mu,\nu)} \int |y - z|^2 \, d\pi(y,z),$$

where $\Pi(\mu,\nu)$ is the set of their couplings. The Brenier map T_μ is the ρ-almost everywhere unique map to μ minimizing the quadratic transport cost. It is a convex gradient. Section 6 also obtains existence and uniqueness in our setting directly from finite approximation and convex duality. Targets may be atomic or singular.

Write

$$\delta = W_2(\mu,\nu), \qquad E = \|T_\mu - T_\nu\|_{L^2(\rho)}, \qquad \tau_\mu(H) = \int_{\{|y|>H\}} |y|^2 \, d\mu(y).$$

For $p \geq 2$ and $\beta \geq 0$, put

$$M_{p,\beta}(\mu) = \int |y|^p [\log(e + |y|)]^\beta \, d\mu(y),$$

and define the one-sided uniform modulus

$$\Omega_{\rho,p,\beta}(\delta) = \sup\left\{ \|T_\mu - T_\nu\|_{L^2(\rho)} : \mu,\nu \in \mathcal{P}_2(\mathbb{R}^d),\ M_{p,\beta}(\mu) \leq 1,\ W_2(\mu,\nu) \leq \delta \right\}. \tag{1.2}$$

There is no higher-moment assumption on ν.

Theorem 1 (One-sided power-logarithmic moments)**.** *Suppose (1.1) holds. If $p > 2$, $\beta \geq 0$, or if $p = 2$, $\beta > 0$, there are constants $C, \delta_0 > 0$, depending only on ρ, p, β, such that*

$$\Omega_{\rho,p,\beta}(\delta) \leq C\,\delta^{(p-2)/(3p-2)} [\log(e/\delta)]^{-2\beta/(3p-2)} \qquad (0 < \delta \leq \delta_0). \tag{1.3}$$

For the single fixed source

$$K_0 = [0,1] \times [-1,1]^{d-1}, \qquad \rho_0 = 2^{-(d-1)} \mathbf{1}_{K_0}\,dx, \tag{1.4}$$

the reverse bound holds with a positive constant depending only on p, β. It is realized, for every sufficiently small δ, by two four-atom targets each satisfying $M_{p,\beta} \leq 1$.

When $p = 2$ and $\beta = 0$, one instead has

$$\frac{1}{20} \leq \Omega_{\rho_0,2,0}(\delta) \leq 2 + \delta \qquad \text{for all sufficiently small } \delta > 0. \tag{1.5}$$

Thus the two exponents in (1.3) are sharp for the stated source class, and already for ρ_0. We do not assert the matching lower bound for each individual nonuniform log-concave source. At $p = 2$, $\beta > 0$, the right side of (1.3) is a constant times $[\log(e/\delta)]^{-\beta/2}$. Equation (1.5) concerns uniform continuity on a moment-bounded class, not continuity at a fixed target.

The upper bound follows from the following more general estimate.

Theorem 2 (Tail estimate)**.** *Under (1.1), there is a finite constant C_ρ such that, for all $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$ and all $H > 0$,*

$$E^2 \le C_\rho\, \delta^{2/3} H^{4/3} + 12\tau_\mu(H) + 16\delta^2. \tag{1.6}$$

Consequently, if μ is supported in a ball of radius $L > 0$ and $\nu \in \mathcal{P}_2(\mathbb{R}^d)$ is arbitrary, then

$$E \le 4\delta + \sqrt{C_\rho}\, L^{2/3} \delta^{1/3}. \tag{1.7}$$

If μ is a point mass, $E = \delta$.

2 The source and a weighted midpoint coupling

Choose $R > 0$ such that $K \subset B(0, R)$. First, f is bounded. Indeed, there is a ball $B \Subset \operatorname{int} K$ and $m > 0$ with $f \ge m$ on B. For $f(x) > 0$, log-concavity gives $f((x+z)/2) \ge \sqrt{m f(x)}$ for $z \in B$. Integration over $(x + B)/2$ yields

$$f(x) \le \frac{2^{2d}}{m|B|^2}. \tag{2.1}$$

Let $M = \sup f < \infty$. For $0 < t < M$, the set $\{f > t\}$ is nonempty, open, and convex, so $K_t = \overline{\{f > t\}}$ is a convex body contained in K. Define

$$P(D) = \sum_{j=1}^{d} |\operatorname{proj}_{e_j^\perp} D|_{d-1}, \qquad Q_\rho = \int_0^M P(K_t)\, dt. \tag{2.2}$$

Then $0 < Q_\rho \le MP(K) < \infty$. The boundary of a convex body has Lebesgue measure zero. Thus the layer-cake formula gives, for every nonnegative Borel g,

$$\int g\, d\rho = \int_0^M \int_{K_t} g(x)\, dx\, dt. \tag{2.3}$$

Lemma 3 (Weighted midpoint domination)**.** *Let $U, V \subset K$ be convex Borel sets of positive ρ-mass α, β. There is a coupling (X, Z) of $\rho|_U/\alpha$ and $\rho|_V/\beta$ for which*

$$\sqrt{\alpha\beta}\, \operatorname{Law}((X+Z)/2) \le \rho \tag{2.4}$$

as measures. The midpoint belongs to $(\overline{U} + \overline{V})/2$ almost surely.

Proof. Replacing each set by its closure changes no integral: a positive-volume convex set has nonempty interior and its boundary has zero volume. We describe the triangular quantile map for a closed convex set $C \subset K$ of mass $a > 0$.

For $0 \le k \le d$, write

$$g_k(x_1, \dots, x_k) = \int_{\mathbb{R}^{d-k}} f(x) \mathbf{1}_C(x)\, dx_{k+1} \cdots dx_d, \qquad g_0 = a.$$

On the interior of the projection of C onto the first k coordinates, g_k is positive and continuous. To justify continuity when $k < d$, fix an interior prefix. Its affine fiber meets $\operatorname{int} C$, and the fiber boundary has zero $(d-k)$-dimensional measure. At the remaining fiber points, membership in C is locally stable as the prefix varies; at interior points f is continuous. Boundedness of f and of C gives dominated convergence. For $k = d$ this is just interior continuity of f on C.

Given an interior prefix $(x_1, \dots, x_{k-1})$, the conditional density

$$t \longmapsto \frac{g_k(x_1, \dots, x_{k-1}, t)}{g_{k-1}(x_1, \dots, x_{k-1})}$$

is positive and continuous on its open support interval and has integral one. Its inverse cumulative distribution is C^1 in the quantile variable in $(0,1)$. Recursively applying these inverse distributions constructs a measurable triangular map $X_C : (0,1)^d \to \operatorname{int} C$. Successive one-variable substitutions show that $(X_C)_{\#} ds = \rho|_C / a$. Its kth component depends only on $s_1, \dots, s_k$, is strictly increasing in s_k, and satisfies

$$\prod_{k=1}^{d} \partial_{s_k} (X_C)_k(s) = \frac{a}{f(X_C(s))}. \tag{2.5}$$

This product telescopes from the conditional-density formulas. No derivatives in the earlier variables are required.

Use the same uniform $s \in (0,1)^d$ for $X = X_U(s)$ and $Z = X_V(s)$, and put $H = (X+Z)/2$. The arithmetic–geometric mean inequality on the positive diagonal derivatives, followed by log-concavity, gives

$$f(H) \prod_{k=1}^{d} \partial_{s_k} H_k \geq \sqrt{\alpha\beta} \frac{f(H)}{\sqrt{f(X) f(Z)}} \geq \sqrt{\alpha\beta}. \tag{2.6}$$

For a nonnegative Borel test function h, substitute first in s_d, then in s_{d-1}, and so on. Each substitution is along a strictly increasing C^1 map with positive derivative. Enlarge its image interval to the whole real line after that substitution. The diagonal factors for earlier variables are independent of the variable being integrated. Tonelli's theorem therefore gives

$$\int_{(0,1)^d} h(H) f(H) \prod_k \partial_{s_k} H_k \, ds \leq \int_{\mathbb{R}^d} h(x) f(x) \, dx.$$

Together with (2.6), this proves (2.4). Endpoint blow-up of quantile derivatives causes no difficulty: each substitution is on an open interval and applies to nonnegative integrands, by monotone convergence from compact subintervals. The support statement follows directly from the construction. □

3 Differentiating weighted cells

For distinct sites $y_1, \dots, y_N \in \mathbb{R}^d$ and intercepts $c_i \in \mathbb{R}$, set

$$u(x) = \max_i \{x \cdot y_i + c_i\}, \qquad P_i = K \cap \bigcap_{j \neq i} \{x : x \cdot (y_i - y_j) + c_i - c_j \geq 0\}.$$

Distinct-site ties lie on finitely many hyperplanes, hence have ρ-measure zero. Put

$$M_i = \int_{P_i} f \, dx, \quad B_i = \int_{P_i} x f \, dx, \quad F_{ij} = P_i \cap P_j,$$

and define symmetric coefficients

$$g_{ij} = \frac{\int_{F_{ij}} f\, dS}{|y_i - y_j|}, \qquad b_{ij} = \frac{\int_{F_{ij}} x f\, dS}{|y_i - y_j|}. \tag{3.1}$$

Here dS is $(d-1)$-dimensional measure on the equality hyperplane; lower-dimensional faces contribute zero.

Proposition 4 (Cell calculus)**.** *On the open set of distinct-site configurations with all $M_i > 0$, masses and moments are jointly C^1 in sites and intercepts. For intercept and site directions $r = (r_i)$ and $v = (v_i)$,*

$$DM_i[r, v] = \sum_{j \neq i} \{g_{ij}(r_i - r_j) + b_{ij} \cdot (v_i - v_j)\}, \tag{3.2}$$

$$DB_i[q, 0] = \sum_{j \neq i} b_{ij}(q_i - q_j). \tag{3.3}$$

The graph with edges $g_{ij} > 0$ is connected. Therefore the symmetric map

$$(Gq)_i = DM_i[q, 0] \tag{3.4}$$

has kernel the constant vectors and range $\{z : \sum_i z_i = 0\}$.

Proof. Each positive-mass cell contains an interior point of K at which its affine function is strictly maximal. Points strictly active for i and j lie on opposite sides of their equality hyperplane H_{ij}, so H_{ij} meets $\operatorname{int} K$. For any affine hyperplane H meeting $\operatorname{int} K$,

$$\operatorname{relint}(K \cap H) = \operatorname{int} K \cap H.$$

Indeed, a relative interior point can be expressed as a strict convex combination of a point in $\operatorname{int} K \cap H$ and a point of $K \cap H$ beyond it. Consequently $\partial K \cap H$ is surface-null. Furthermore, H_{ij} and H_{ik} cannot coincide for distinct indices. Otherwise the three affine functions lie on an affine line and one is a strict convex combination of the other two, contradicting its strict activity. Triple equality sets thus have dimension at most $d - 2$.

For $\varphi = 1$ or a coordinate function, express the weighted cell integral as

$$I_i^{\varphi} = \int f(x)\varphi(x)\mathbf{1}_K(x) \prod_{j \neq i} \mathbf{1}_{\{\ell_{ij}(x) \geq 0\}}\, dx, \qquad \ell_{ij} = x \cdot (y_i - y_j) + c_i - c_j.$$

Perturb by $h(r, v)$, and telescope the product of indicators one factor at a time. The term in which factor j changes is supported in $|\ell_{ij}| \leq C|h|$ on a common bounded region. With $n_{ij} = (y_i - y_j)/|y_i - y_j|$, use $x = z + htn_{ij}$ for $z \in H_{ij}$ and initially $h > 0$. The volume factor h cancels the difference quotient. Away from the surface-null boundary and triple-equality sets, the other indicators converge to their values at z, and the changed factor tends to

$$\mathbf{1}_{\{|y_i - y_j|t + r_i - r_j + z \cdot (v_i - v_j) \geq 0\}} - \mathbf{1}_{\{t \geq 0\}}.$$

Its integral in t is $(r_i - r_j + z \cdot (v_i - v_j))/|y_i - y_j|$. The density is bounded and continuous at the interior points in question. Dominated convergence on the bounded (z, t) region gives

$$DI_i^{\varphi}[r, v] = \sum_{j \neq i} \frac{1}{|y_i - y_j|} \int_{F_{ij}} \varphi(x) f(x)(r_i - r_j + x \cdot (v_i - v_j))\, dS(x). \tag{3.5}$$

Applying the argument in the opposite direction gives the two-sided derivative. Choosing $\varphi = 1$ and coordinates proves the displayed formulas.

The coefficients are continuous as the configuration varies. Identify a nearby equality hyperplane with H_{ij} by an affine graph in direction n_{ij}. Its graph map and surface Jacobian converge uniformly on a common bounded region. Outside the same null boundary and triple-equality sets, face indicators and weighted integrands converge. Denominators stay positive locally. Dominated convergence applies also to the quadratic coordinate factors arising from site derivatives of B_i. All coordinate partial derivatives in (3.5) are therefore continuous. The configuration set is open since masses are continuous by dominated convergence away from the tie hyperplanes.

For connectivity, choose strictly active interior points in any two cells so that their joining segment avoids every triple equality set. This is possible by excluding finitely many proper affine sets from the open active regions. The segment lies inside $\operatorname{int} K$. At a change of active index it crosses a face with a relatively open disk in $\operatorname{int} K$, and f is positive there. This face gives $g_{ij} > 0$. The successive indices form a graph path. Finally,

$$q \cdot Gq = \sum_{i<j} g_{ij}(q_i - q_j)^2.$$

Connectivity proves the kernel statement, and symmetry proves the range statement. □

Proposition 5 (Prescribed positive masses)**.** *For positive m_i with $\sum_i m_i = 1$ and any distinct sites, there is a unique intercept vector satisfying*

$$M_i = m_i, \qquad \sum_i m_i c_i = 0. \tag{3.6}$$

With m fixed, this vector depends C^1 on the sites.

Proof. For $N = 1$ take $c_1 = 0$. Otherwise minimize

$$\mathcal{F}_y(c) = \int \max_i (x \cdot y_i + c_i)\, d\rho - \sum_i m_i c_i$$

on $H_m = \{c : \sum_i m_i c_i = 0\}$. If $S = \max_i |y_i|$, then $\mathcal{F}_y(c) \geq \max_i c_i - RS$ on H_m. An upper bound on all c_i, together with positive weights and their weighted mean zero, bounds every c_i below. Thus this continuous convex function is coercive on H_m and attains a minimum.

Almost-everywhere uniqueness of the active index and bounded difference quotients give $\nabla_c \mathcal{F}_y = (M_i - m_i)_i$. At a constrained minimum this vector is a multiple of m. Its coordinate sum is zero, so the multiple is zero. This proves existence of (3.6), and every solution is a minimizer. At a solution, Proposition 4 shows that

$$G : H_m \longrightarrow H_0, \qquad H_0 = \{z : \sum_i z_i = 0\},$$

is an isomorphism. The implicit function theorem gives a locally unique C^1 branch. Two normalized solutions would have an entire segment of minimizers between them, contradicting local isolation. The local branches therefore agree, proving uniqueness and the claimed global C^1 dependence. No quantitative bound on G^{-1} is used. □

4 A uniform mass–moment comparison

Proposition 6. *At every configuration in Proposition 4, and for every intercept direction q,*

$$\sum_i \frac{|DB_i[q,0]|^2}{M_i} \le 162R^2 \sum_i \frac{|DM_i[q,0]|^2}{M_i}. \tag{4.1}$$

The constant is independent of sites, intercepts, atom count, and the positive masses.

Proof. Keep the sites fixed and perturb intercepts to $c \pm sq$. Write P_i^-, P_i^+ for the cells, α_i, β_i for their masses, $\gamma_i = \sqrt{\alpha_i\beta_i}$, and $\Delta_i = \alpha_i - \beta_i$. Apply Lemma 3 to obtain a coupling (X_i, Z_i) of the normalized restrictions. Averaging the inequalities defining P_i^- and P_i^+ shows that their midpoint lies in P_i. Hence

$$\gamma_i \,\mathrm{Law}((X_i + Z_i)/2) \le \rho|_{P_i}.$$

Put $D_i = \mathbb{E}|X_i - Z_i|^2$, $\Phi(x) = R^2 - |x|^2$, and $A_i^- = \mathbb{E}\Phi(X_i)$, $A_i^+ = \mathbb{E}\Phi(Z_i)$. Testing midpoint domination with the nonnegative function Φ and summing gives

$$\sum_i \gamma_i \left\{ \frac{A_i^- + A_i^+}{2} + \frac{D_i}{4} \right\} \le \int \Phi \, d\rho = \frac{1}{2} \sum_i (\alpha_i A_i^- + \beta_i A_i^+).$$

Here all three cell families partition the source. We have $0 \le A_i^\pm \le R^2$, $|A_i^- - A_i^+| \le 2R\sqrt{D_i}$, and

$$\frac{\alpha_i + \beta_i}{2} - \gamma_i = \frac{(\sqrt{\alpha_i} - \sqrt{\beta_i})^2}{2} \le \frac{\Delta_i^2}{8\gamma_i}.$$

For $\mathcal{A} = \sum_i \gamma_i D_i$ and $\mathcal{B} = \sum_i \Delta_i^2/\gamma_i$, the preceding identities and Cauchy–Schwarz give

$$\frac{\mathcal{A}}{4} \le \frac{R^2\mathcal{B}}{8} + \frac{R}{2}\sqrt{\mathcal{A}\mathcal{B}} \le \frac{R^2\mathcal{B}}{8} + \frac{\mathcal{A}}{8} + \frac{R^2\mathcal{B}}{2}.$$

Thus $\mathcal{A} \le 5R^2\mathcal{B}$.

The perturbed moment difference is $\Delta B_i = \alpha_i \mathbb{E}X_i - \beta_i \mathbb{E}Z_i$, whence

$$|\Delta B_i| \le R|\Delta_i| + \alpha_i\sqrt{D_i}.$$

For all sufficiently small s, continuity of the masses ensures $\alpha_i, \beta_i, \gamma_i \in [M_i/2, 2M_i]$ for every i. Set $\mathcal{E} = \sum_i \Delta_i^2/M_i$. Then $\alpha_i^2/M_i \le 8\gamma_i$ and $\mathcal{B} \le 2\mathcal{E}$. Consequently,

$$\sum_i \frac{|\Delta B_i|^2}{M_i} \le 2R^2\mathcal{E} + 16\mathcal{A} \le (2 + 160)R^2\mathcal{E}.$$

Divide by $(2s)^2$ and let $s \to 0$, using Proposition 4. The smallness threshold for s may depend on the configuration. The constant in the limiting estimate does not. □

5 Radius-independent stability of finite potentials

We first record optimality and normalization. If u is a finite maximum of affine functions and $\lambda = (\nabla u)_\# \rho$, then the Fenchel inequality

$$x \cdot y \le u(x) + u^*(y)$$

is an equality on $y = \nabla u(x)$ wherever u is differentiable. Both dual terms are integrable for this finite law. Integrating proves that this graph maximizes correlation, equivalently minimizes quadratic cost, among all couplings of ρ and λ. Any other optimal coupling must also have equality almost everywhere. At a differentiability point $\partial u(x) = \{\nabla u(x)\}$, so the optimal coupling is uniquely the graph coupling.

Two convex potentials with the same gradient ρ-almost everywhere on $\operatorname{int} K$ differ by a constant there. Indeed, $f > 0$ on the interior, so their locally Lipschitz difference has zero gradient Lebesgue-almost everywhere. Mollifying on an interior ball makes that difference constant on smaller balls; connectedness then gives a single constant. Normalization by $\int u \, d\rho = 0$ therefore determines the potential as an element of $L^2(\rho)$.

Proposition 7. *Let μ, ν be finitely supported probabilities. Their normalized convex potentials satisfy*

$$\|u_\mu - u_\nu\|_{L^2(\rho)} \le A_\rho W_2(\mu, \nu), \qquad A_\rho = (1 + \sqrt{162})R. \tag{5.1}$$

No bound on the target supports occurs in A_ρ.

Proof. The finite coupling polytope is nonempty and compact, so a minimum-cost coupling exists. Discard its zero weights, merge identical pairs, and write it as

$$\pi = \sum_{i=1}^{N} m_i \delta_{(y_i^0, y_i^1)}, \qquad v_i = y_i^1 - y_i^0, \qquad y_i(t) = (1-t)y_i^0 + t y_i^1.$$

Then $m_i > 0$, $\sum_i m_i = 1$ and $\sum_i m_i |v_i|^2 = \delta^2$. Swapping equal positive portions of two coupling masses shows that

$$(y_i^0 - y_j^0) \cdot (y_i^1 - y_j^1) \ge 0.$$

If $a = y_i^0 - y_j^0$ and $b = y_i^1 - y_j^1$, it follows that

$$|(1-t)a + tb|^2 = (1-t)^2|a|^2 + t^2|b|^2 + 2t(1-t)a \cdot b > 0$$

for $0 < t < 1$, because the two endpoint pairs are distinct. Thus the interpolated sites are distinct at every interior time, even when endpoint sites coincide.

Use Proposition 5 to choose $c_i(t)$ with cell masses m_i and $\sum_i m_i c_i(t) = 0$. Put $r_i = c_i'(t)$. Differentiating the mass constraints and using the symmetry in (3.2)–(3.3), we obtain for every q

$$\sum_i r_i DM_i[q, 0] = -\sum_i v_i \cdot DB_i[q, 0]. \tag{5.2}$$

For clarity, the site term after multiplying the mass equation by q_i and summing is $\sum_{i<j}(q_i - q_j)b_{ij} \cdot (v_i - v_j) = \sum_i v_i \cdot DB_i[q, 0]$. Since $\sum_i m_i r_i = 0$, Proposition 4 permits a choice of q with $DM_i[q, 0] = m_i r_i$. With $E_r = \sum_i m_i r_i^2$, equation (5.2) and Proposition 6 imply

$$E_r \le \left(\sum_i m_i |v_i|^2 \right)^{1/2} \left(\sum_i \frac{|DB_i[q, 0]|^2}{m_i} \right)^{1/2} \le \sqrt{162} R\delta \sqrt{E_r}.$$

Hence $\|r\|_{\ell^2(m)} \le \sqrt{162} R\delta$, including when $E_r = 0$. For $N = 1$ this holds directly with $r_1 = 0$.

Let $u_t(x) = \max_i (x \cdot y_i(t) + c_i(t))$. On compact subintervals of $(0, 1)$ it is uniformly Lipschitz in time for $x \in K$. Away from ties its time derivative is $x \cdot v_i + r_i$ on cell i. Ties have zero source measure at each time, so Fubini gives the derivative almost everywhere in (t, x). Thus

$$\|\partial_t u_t\|_{L^2(\rho)} \le R \|v\|_{\ell^2(m)} + \|r\|_{\ell^2(m)} \le A_\rho \delta.$$

The fundamental theorem of calculus and Minkowski's inequality give

$$\|u_t - u_s\|_{L^2(\rho)} \le A_\rho \delta |t-s| \qquad (0 < s, t < 1). \tag{5.3}$$

Each $|c_i'(t)| \le \sqrt{162} R\delta/\sqrt{m_i}$, so every intercept has finite limits at both endpoints. The maxima converge uniformly on K to endpoint maxima. At either endpoint merge repeated slopes by retaining their largest intercept. Except on finitely many hyperplanes the maximizing slope is unique, and all active slopes at approaching interior times tend to it: otherwise a subsequence of a fixed active index would contradict the strict endpoint gap. The slopes are bounded for this fixed finite pair. Bounded convergence against continuous tests shows that the endpoint gradient laws are μ and ν. The optimality argument preceding the proposition identifies the endpoint maps.

Take endpoint limits in (5.3). Finally replace each endpoint potential by its ρ-mean-zero version. Orthogonal projection onto the mean-zero subspace of $L^2(\rho)$ cannot increase their difference norm. This proves (5.1). The mass-dependent intercept estimate was used only for endpoint existence, not for the final constant. □

6 Extension to arbitrary finite-second-moment targets

Lemma 8 (Local compactness of convex functions)**.** *Let $O \subset \mathbb{R}^d$ be open and convex. A sequence of finite convex functions with uniformly bounded L^1 norms on every compact subset of O has a subsequence converging locally uniformly to a finite convex function. If $u_n \to u$ locally uniformly, then $\nabla u_n(x) \to \nabla u(x)$ at every point where u and all u_n are differentiable.*

Proof. We give the needed local bounds. On a ball with its closure inside O, Jensen's inequality bounds $u_n(z)$ from above by its average on a fixed smaller ball centered at z. An L^1 bound on a larger concentric ball therefore gives a uniform upper bound U on an intermediate ball. For z in an inner ball and y in a ball $B(z,r)$ contained in that intermediate ball, convexity gives

$$u_n((z+y)/2) \le \tfrac{1}{2}(u_n(z) + U).$$

If $u_n(z) + U < 0$, integration of the absolute value on $B(z, r/2)$ bounds $-u_n(z) - U$ by the larger-ball L^1 norm divided by a fixed positive volume. This proves a uniform lower bound on the inner ball. Difference quotients along line segments, using the two-sided bounds on a slightly larger ball, give a uniform Lipschitz constant on a still smaller ball. Finite coverings and Arzelà–Ascoli, followed by a countable compact exhaustion, yield a locally uniform subsequential limit, which is finite and convex.

At a common differentiability point, convexity gives for each coordinate and small $h > 0$

$$\frac{u_n(x) - u_n(x - he_j)}{h} \le \partial_j u_n(x) \le \frac{u_n(x + he_j) - u_n(x)}{h}.$$

First let $n \to \infty$, then $h \downarrow 0$. Both bounds tend to $\partial_j u(x)$. □

Proposition 9. *For every $\lambda \in \mathcal{P}_2(\mathbb{R}^d)$ there is a convex function u_λ on $\operatorname{int} K$, belonging to $L^2(\rho)$ and normalized by $\int u_\lambda \, d\rho = 0$, whose gradient is the unique quadratic optimal map to λ. Inequality (5.1) holds for all $\mu, \nu \in \mathcal{P}_2(\mathbb{R}^d)$. If finite targets $\lambda_n \to \lambda$ in W_2, their normalized potentials converge in $L^2(\rho)$ and locally uniformly on $\operatorname{int} K$, and their gradients converge in $L^2(\rho)$.*

Proof. Finite targets are dense in $\mathcal{P}_2(\mathbb{R}^d)$: move the mass outside a growing ball to zero, then quantize the mass inside that ball on a grid of shrinking mesh. The resulting deterministic couplings have squared displacement tending to zero. Choose such a sequence λ_n. Proposition 7 makes the corresponding normalized potentials Cauchy in $L^2(\rho)$.

On every compact subset of $\operatorname{int} K$, f has a positive lower bound. The $L^2(\rho)$ bound therefore gives a local $L^1(dx)$ bound. Lemma 8 produces a locally uniform convex limit along a subsequence. It agrees almost everywhere with the $L^2(\rho)$ limit. Any two such continuous convex limits agree on the interior, since f is positive there. Applying this compactness argument to every subsequence proves local uniform convergence of the whole sequence to a finite convex function u. Its ρ-mean is zero, and $u \in L^2(\rho)$.

Convex functions are locally Lipschitz on their open domain and hence differentiable almost everywhere. Remove the null sets for u and the countably many u_n. Lemma 8 gives $\nabla u_n \to \nabla u$ at all remaining interior points. The boundary carries no ρ-mass. For a bounded continuous test function, dominated convergence and weak convergence $\lambda_n \to \lambda$ show

$$(\nabla u)_{\#}\rho = \lambda.$$

We also justify strong L^2 convergence of the gradients. Whenever two random vectors Y, Z are coupled, elementary splitting into $|Y| \le H$ and $|Y| > H$ gives

$$\mathbb{E}[|Z|^2 \mathbf{1}_{\{|Z|>2H\}}] \le 4\mathbb{E}|Z-Y|^2 + 2\mathbb{E}[|Y|^2 \mathbf{1}_{\{|Y|>H\}}]. \tag{6.1}$$

On the first event, $|Z| \le 2|Z-Y|$; on the second, use $|Z|^2 \le 2|Y|^2 + 2|Z-Y|^2$. Couplings with laws λ, λ_n and squared displacement tending to zero show that the second-moment tails of λ_n are uniformly small: first handle all sufficiently large n using (6.1), and then the finitely many remaining n individually. Thus $|\nabla u_n|^2$ is uniformly integrable on the fixed source space. The almost-everywhere convergence and Vitali's theorem imply $\nabla u_n \to \nabla u$ in $L^2(\rho)$.

To identify optimality without any bounded-potential assumption at the boundary, extend u as a proper lower semicontinuous convex function on $\mathbb{R}^d$ by taking the closure of its epigraph over $\operatorname{int} K$ and assigning $+\infty$ outside K. A supporting affine function at an interior point bounds it below, so this closure is proper and agrees with u on the interior. At differentiability points,

$$u^*(\nabla u(x)) = x \cdot \nabla u(x) - u(x).$$

The right side is absolutely integrable because x is bounded, $\nabla u \in L^2(\rho)$, and $u \in L^2(\rho)$. Therefore $u^* \in L^1(\lambda)$. Fenchel's inequality can now be integrated against every coupling of ρ and λ; both dual terms are integrable and correlation is integrable by the second moments. Equality on the graph proves optimality. Every optimal coupling has zero integrated nonnegative Fenchel gap and hence equality almost everywhere. Differentiability of u forces it to be the same graph coupling. This proves uniqueness. The earlier gradient-uniqueness argument also proves uniqueness of the normalized potential.

For arbitrary μ, ν, choose finite approximations and pass to the limit in (5.1), using their $L^2(\rho)$ convergence and $W_2(\mu_n, \nu_n) \to W_2(\mu, \nu)$. The same argument applies to any finite approximating sequence, proving the final assertion. □

7 Convex-gradient interpolation with tails

The next estimate uses one-dimensional clipping only inside an inequality. It does not require a radial truncation of a Brenier map to remain a convex gradient.

Lemma 10 (A clipped secant estimate)**.** *Let u, v be finite convex functions on an open interval, let $h > 0$, and suppose the derivatives at x and $x + h$ exist. Write $w = u - v$ and $D_h w(x) = (w(x + h) - w(x))/h$. If $|u'(x)|, |v'(x)| \le H$, then*

$$|u'(x) - v'(x)| \le |D_h w(x)| + u'_H(x + h) - u'_H(x) + v'_H(x + h) - v'_H(x), \tag{7.1}$$

where $a_H = \max(-H, \min(a, H))$ denotes clipping to $[-H, H]$.

Proof. Suppose $a = u'(x) \ge b = v'(x)$. Convexity gives $a \le D_h u(x)$ and $D_h v(x) \le v'(x + h)$. If $v'(x + h) \le H$, monotonicity and $b \ge -H$ imply

$$a - b \le D_h w(x) + v'(x + h) - b \le |D_h w(x)| + v'_H(x + h) - v'_H(x).$$

If $v'(x + h) > H$, then $a - b \le H - b = v'_H(x + h) - v'_H(x)$. The omitted clipped increment of u' is nonnegative. If $a < b$, interchange u, v. □

Proposition 11 (Weighted interpolation)**.** *Let u, v be finite convex functions on $\operatorname{int} K$, with $u - v \in L^2(\rho)$ and $\nabla u, \nabla v \in L^2(\rho)$. Put $\mu = (\nabla u)_{\#}\rho$, $\nu = (\nabla v)_{\#}\rho$. For every $h, H > 0$,*

$$\|\nabla u - \nabla v\|^2_{L^2(\rho)} \le 12dh^{-2} \|u - v\|^2_{L^2(\rho)} + 28Q_\rho H^2 h + 4\{\tau_\mu(H) + \tau_\nu(H)\}. \tag{7.2}$$

Proof. First work with unweighted Lebesgue measure on the interior of a convex body $D \subset K$ where $w = u - v$ is square-integrable. Let

$$G = \{x : |\nabla u(x)| \le H,\ |\nabla v(x)| \le H\},$$

disregarding the null set of nondifferentiability. Fix a coordinate direction e_j. Almost every line parallel to it meets the interior of D in an interval $I = (a, b)$. On the portion where $x, x + he_j \in \operatorname{int} D$, apply Lemma 10. The one-variable derivatives have monotone representatives; their clipped versions g are monotone and bounded by H. Direct translation of integrals gives

$$\int_a^{b-h} (g(t + h) - g(t))\, dt \le 2Hh,$$

when $b - a > h$, and the integral is zero otherwise. Since the increment is between 0 and $2H$, its squared integral is at most $4H^2h$.

Square (7.1) using $(A + B + C)^2 \le 3(A^2 + B^2 + C^2)$ and integrate over G in the permissible region. The secant term is bounded by

$$3h^{-2} \int_{D \cap (D - he_j)} |w(x + he_j) - w(x)|^2\, dx \le 12h^{-2} \int_D |w|^2\, dx.$$

Each of the two clipped increments contributes at most $12H^2h|\operatorname{proj}_{e_j^\perp} D|_{d-1}$. At the missing end of each interval the length is at most h, even if the entire interval is shorter than h; on G, the squared coordinate difference is at most $4H^2$. Hence

$$\int_{G \cap D} |\partial_j u - \partial_j v|^2\, dx \le 12h^{-2} \int_D |w|^2\, dx + 28H^2h|\operatorname{proj}_{e_j^\perp} D|_{d-1}.$$

Fubini justifies the linewise calculation; translating the exceptional derivative set by he_j still gives a null set. The derivatives need not be bounded before clipping.

Sum over j. Outside G one has the pointwise estimate

$$|\nabla u - \nabla v|^2 \mathbf{1}_{G^c} \le 4(|\nabla u|^2 \mathbf{1}_{\{|\nabla u|>H\}} + |\nabla v|^2 \mathbf{1}_{\{|\nabla v|>H\}}). \tag{7.3}$$

If only one gradient exceeds H, its norm dominates the other; if both do, the usual squared triangle inequality suffices.

Apply the unweighted estimate on $\operatorname{int} K_t$ and integrate in t. Equation (2.3) gives all weighted integrals and (2.2) gives the projection term. For almost every t the required square-integrability of w holds by (2.3) and Tonelli. All evaluations and secants occur in the interiors; boundary values of the potentials are unnecessary. This also covers potentials diverging near ∂K. Finally integrate (7.3) against ρ to obtain the tail terms. □

Proof of Theorem 2. Apply Propositions 9 and 11. With

$$A_0 = 12dA_\rho^2, \qquad B_0 = 28Q_\rho,$$

we obtain

$$E^2 \le A_0\delta^2 h^{-2} + B_0 H^2 h + 4\{\tau_\mu(H) + \tau_\nu(H)\}.$$

If $\delta = 0$, uniqueness gives $E = 0$. Otherwise take $h = (2A_0\delta^2/(B_0H^2))^{1/3}$. There is no upper restriction on h in Proposition 11. This gives

$$E^2 \le C_0\delta^{2/3}H^{4/3} + 4\{\tau_\mu(H) + \tau_\nu(H)\}, \qquad C_0 = 3\,2^{-2/3}A_0^{1/3}B_0^{2/3}. \tag{7.4}$$

Applying (6.1) to an optimal coupling, or to couplings with costs tending to the infimum, yields

$$\tau_\nu(2H) \le 4\delta^2 + 2\tau_\mu(H).$$

Use threshold $2H$ in (7.4), and note $\tau_\mu(2H) \le \tau_\mu(H)$. This proves (1.6) with the explicit choice

$$C_\rho = 2^{4/3}\,3\,2^{-2/3}(12dA_\rho^2)^{1/3}(28Q_\rho)^{2/3}. \tag{7.5}$$

If μ is supported in $B(0, L)$, choose $H = L$. For a ball centered elsewhere, translate both targets by its center. Their maps translate by the same vector: this preserves pushforward and changes every admissible quadratic cost by the same quantity. Both E and δ are unchanged. Taking square roots proves (1.7). For a point target z, $T_\mu = z$ almost everywhere and both E^2 and δ^2 equal $\int |y - z|^2\, d\nu(y)$. □

8 Optimization under a moment constraint

Upper bound in Theorem 1. Let $M_{p,\beta}(\mu) \le 1$. For $H \ge 1$, monotonicity of $r^{p-2}[\log(e + r)]^\beta$ gives

$$\tau_\mu(H) \le H^{2-p}[\log(e + H)]^{-\beta}. \tag{8.1}$$

Suppose first $0 < \delta < 1$ is the actual distance of the two targets. Write

$$D = 3p - 2, \qquad \ell = \log(e/\delta), \qquad H = \delta^{-2/D}\ell^{-3\beta/D}.$$

For sufficiently small δ, $H \ge 1$, and

$$\log H = \frac{2}{D}\log(1/\delta) - \frac{3\beta}{D}\log \ell$$

shows that $\log(e+H)$ is bounded above and below by positive multiples of ℓ, with constants depending only on p, β. Straight calculation yields

$$\delta^{2/3} H^{4/3} = \delta^{2(p-2)/D} \ell^{-4\beta/D},$$
$$H^{2-p}[\log(e+H)]^{-\beta} \le C_{p,\beta} \delta^{2(p-2)/D} \ell^{-4\beta/D}.$$

The second line uses $3\beta(p-2)/D - \beta = -4\beta/D$. The term δ^2 is bounded by a constant times the same expression for all sufficiently small δ. Theorem 2 and a square root prove (1.3) for the actual distance. This argument applies directly at $p = 2$, $\beta > 0$, yielding $E \le C\ell^{-\beta/2}$.

To pass to the supremum in (1.2), the function $s^{(p-2)/D}[\log(e/s)]^{-2\beta/D}$ is increasing for $0 < s < 1$: its logarithmic derivative is

$$\frac{1}{s}\left(\frac{p-2}{D} + \frac{2\beta}{D\log(e/s)}\right) > 0$$

under the present assumptions. Zero distance gives zero error by uniqueness. Thus the same bound holds for all pairs at distance at most δ.

For $p = 2$, $\beta = 0$, $\|T_\mu\|_2 \le 1$. Minkowski's inequality in a coupling shows that the root second moment of ν is at most $1 + \delta$. Therefore $E \le 2 + \delta$, proving the upper half of (1.5). □

9 Four atoms on a fixed source: sharpness and the endpoint

All calculations in this section use the fixed source (1.4). Only the first two coordinates matter; write $t = x_1$, $s = x_2$. Let $0 < a < 1/6$, $R_* > 0$ and $b = R_* a/4$. The symbol R_* here is a target scale and is unrelated to the enclosing source radius R. Define

$$u(t,s) = R_* \max\{0, t - 1 + 3a, 2t - 2 + 5a, 3t - 3 + 6a\}, \tag{9.1}$$
$$v(t,s) = \max\{0, R_*(t - 1 + 3a), R_*(2t - 2 + 5a) + bs, R_*(3t - 3 + 6a)\}. \tag{9.2}$$

Proposition 12 (Exact transport and error formulas)**.** *The gradient laws of* (9.1)–(9.2) *are*

$$\mu_a = (1-3a)\delta_0 + a\delta_{R_* e_1} + a\delta_{2R_* e_1} + a\delta_{3R_* e_1}, \tag{9.3}$$
$$\nu_a = (1-3a)\delta_0 + a\delta_{R_* e_1} + a\delta_{2R_* e_1 + b e_2} + a\delta_{3R_* e_1}. \tag{9.4}$$

Their optimal maps are ∇u and ∇v, and

$$\delta_a^2 := W_2(\mu_a, \nu_a)^2 = \frac{R_*^2 a^3}{16}, \qquad E_a^2 := \|T_{\mu_a} - T_{\nu_a}\|^2_{L^2(\rho_0)} = \frac{R_*^2 a}{4} + \frac{R_*^2 a^3}{16}. \tag{9.5}$$

Proof. The slopes in t are $0, R_*, 2R_*, 3R_*$. For u, consecutive intersections are $1-3a, 1-2a, 1-a$. For v they are

$$q_1 = 1 - 3a, \qquad q_2 = 1 - 2a - \tfrac{a}{4}s, \qquad q_3 = 1 - a + \tfrac{a}{4}s.$$

For every $-1 \le s \le 1$ these lie strictly in $(0,1)$ in the stated order. Ordered intersections of consecutive lines with strictly increasing slopes determine their upper envelope: on the interval between consecutive intersections, comparisons with every earlier or later line follow by summing the adjacent comparisons. Thus there are no additional cells from nonadjacent lines.

The four t-widths for v are

$$1 - 3a, \qquad a - \tfrac{a}{4}s, \qquad a + \tfrac{a}{2}s, \qquad a - \tfrac{a}{4}s.$$

Since s is uniform on $[-1,1]$ and has mean zero, both potentials have cell masses $1-3a, a, a, a$. This proves (9.3)–(9.4). The convex-gradient optimality argument in Section 5 identifies the maps.

The coupling fixing three atoms and moving only $2R_*e_1$ to $2R_*e_1+be_2$ has cost ab^2. Conversely every source atom of μ_a has second coordinate zero, while the second-coordinate second moment of ν_a is ab^2. Every coupling has at least this coordinate cost. Hence $\delta_a^2 = ab^2$ exactly.

For each s, the two moving cell boundaries displace by $a|s|/4$ each. Their displacement intervals are disjoint, and they exchange only adjacent slopes; their separation is at least $a/2$. On these intervals the first-coordinate gradient difference has magnitude R_*. Thus the total source mass of first-coordinate change is

$$\frac{a}{2}\mathbb{E}|s| = \frac{a}{4}.$$

The second-coordinate gradient difference equals b precisely on the third cell of v, whose mass is a. Orthogonality of the coordinates gives $E_a^2 = R_*^2 a/4 + ab^2$, proving (9.5). Integrating the unused coordinates contributes one. □

Lower bounds in Theorem 1. Fix $p \geq 2$, $\beta \geq 0$ and set

$$\ell_a = \log(1/a), \qquad R_* = \frac{1}{10}a^{-1/p}\ell_a^{-\beta/p}. \tag{9.6}$$

As $a \downarrow 0$, $R_* \to \infty$ and $e + 3R_* \leq 1/a$ for all sufficiently small a. All nonzero atom norms in (9.3)–(9.4) are at most $3R_*$, since $b \leq R_*$ and $\sqrt{4R_*^2 + b^2} < 3R_*$. Hence, for each target,

$$M_{p,\beta} \leq 3a(3R_*)^p[\log(e+3R_*)]^\beta \leq 3(3/10)^p \leq 27/100 < 1. \tag{9.7}$$

Equation (9.5) becomes

$$\delta_a = \frac{1}{40}a^{3/2-1/p}\ell_a^{-\beta/p}, \qquad E_a = \frac{1}{20}a^{1/2-1/p}\ell_a^{-\beta/p}\sqrt{1+a^2/4}. \tag{9.8}$$

For $\alpha = (p-2)/(3p-2)$,

$$\frac{1}{2} - \frac{1}{p} = \alpha\left(\frac{3}{2} - \frac{1}{p}\right), \qquad -\frac{\beta}{p} = -\frac{\alpha\beta}{p} - \frac{2\beta}{3p-2}.$$

Moreover,

$$\log(e/\delta_a) = \log(40e) + (3/2 - 1/p)\ell_a + (\beta/p)\log\ell_a$$

is comparable to ℓ_a. Therefore, when $p > 2$, $\beta \geq 0$, or $p = 2$, $\beta > 0$,

$$E_a \asymp_{p,\beta} \delta_a^{(p-2)/(3p-2)}[\log(e/\delta_a)]^{-2\beta/(3p-2)}. \tag{9.9}$$

This gives the lower bound at every sufficiently small distance, not merely along a sequence. Indeed, the function $a \mapsto \delta_a$ is continuous, tends to zero, and is strictly increasing on a small interval: its logarithmic derivative is

$$\frac{1}{a}\left(\frac{3}{2} - \frac{1}{p} + \frac{\beta}{p\ell_a}\right) > 0.$$

Choose the upper end of that interval so that (9.7) and all cell conditions hold. The intermediate value theorem then realizes each smaller positive δ exactly.

Finally, when $p = 2$, $\beta = 0$, formula (9.8) yields

$$\delta_a = a/40 \longrightarrow 0, \qquad E_a^2 = \frac{1}{400} + \frac{a^2}{1600} \geq \frac{1}{400}.$$

Both second moments satisfy (9.7). This proves the lower half of (1.5). The mass escapes to increasingly distant target atoms with a nonvanishing contribution to their second moments; their second-moment tails are not uniformly integrable. There is no conflict with the pointwise continuity proved in Proposition 9. □

10 Conclusion

For every fixed compact log-concave source ρ, the normalized-potential estimate and the clipped-gradient interpolation give, for every $H > 0$,

$$\|T_\mu - T_\nu\|_{L^2(\rho)}^2 \leq C_\rho \delta^{2/3} H^{4/3} + 12\tau_\mu(H) + 16\delta^2.$$

Under $M_{p,\beta}(\mu) \leq 1$, optimizing H proves the upper bound in Theorem 1. On the single fixed source ρ_0 in (1.4), the four-atom construction proves the reverse bound. Hence, for $p > 2$, $\beta \geq 0$, or $p = 2$, $\beta > 0$,

$$\Omega_{\rho_0,p,\beta}(\delta) \asymp_{\rho_0,p,\beta} \delta^{(p-2)/(3p-2)} [\log(e/\delta)]^{-2\beta/(3p-2)} \qquad (\delta \downarrow 0).$$

At $p = 2$, $\beta > 0$, this is $[\log(e/\delta)]^{-\beta/2}$. At $p = 2$, $\beta = 0$, the explicit error remains at least $1/20$ while the target distance tends to zero, so no vanishing uniform modulus exists under a second-moment bound alone. These prove all assertions of Theorem 1.